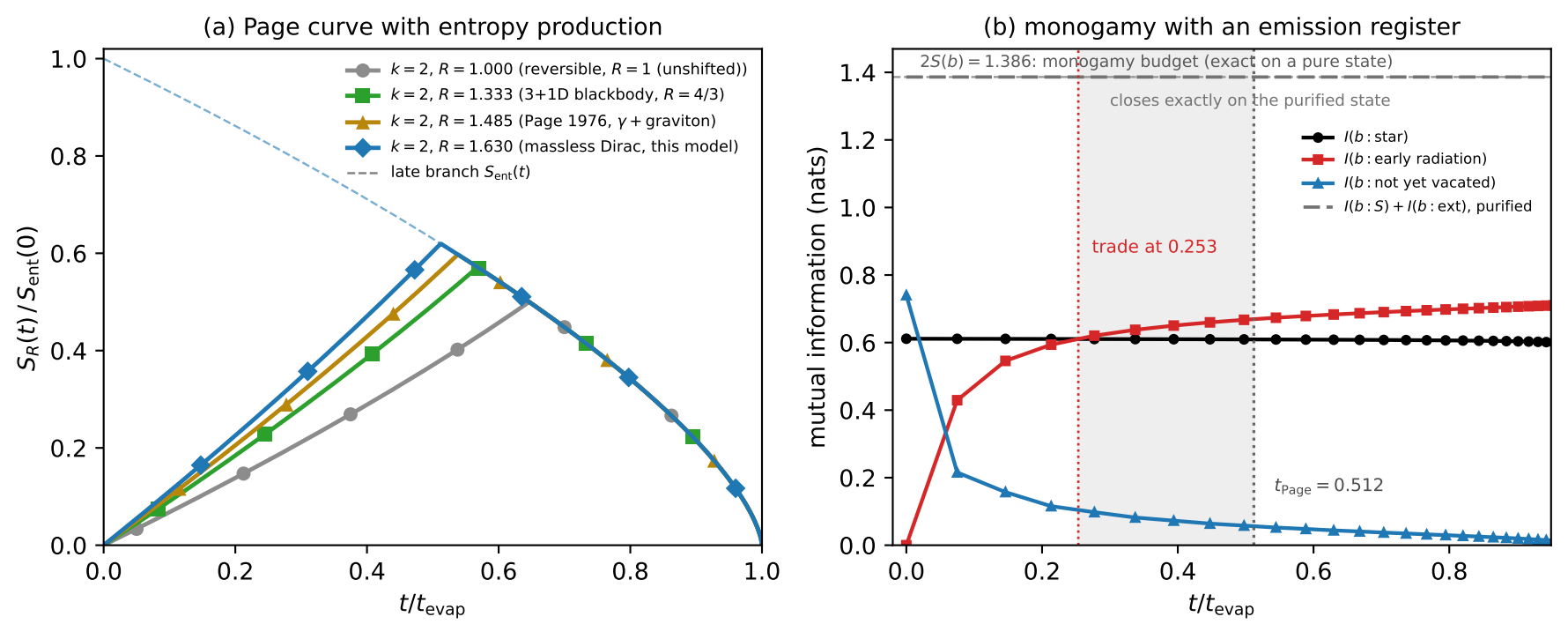
(a) Page curve with entropy production
$k = 2$, $R = 1.000$ (reversible, $R = 1$ (unshifted))
$k = 2$, $R = 1.333$ (3+1D blackbody, $R = 4/3$)
$k = 2$, $R = 1.485$ (Page 1976, γ + graviton)
$k = 2$, $R = 1.630$ (massless Dirac, this model)
late branch $S_{\text{ent}}(t)$
$S_R(t)/S_{\text{ent}}(0)$
t/t_{evap}
(b) monogamy with an emission register
$2S(b) = 1.386$: monogamy budget (exact on a pure state)
closes exactly on the purified state
$I(b:\text{star})$
$I(b:\text{early radiation})$
$I(b:\text{not yet vacated})$
$I(b:S) + I(b:\text{ext})$, purified
trade at 0.253
$t_{\text{Page}} = 0.512$
mutual information (nats)
t/t_{evap}